MIL-P-60447A (MU)
15 March 1971
SUPERSEDING
MIL-P-60447
28 May 1968

MILITARY SPECIFICATION

# PROJECTILE, 81MM, ILLUMINATING, M301A3 PARTS LOADING, ASSEMBLING AND PACKING

## 1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of mortar projectile designated as Projectile, 81MM, Illuminating, M301A3.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 - Ammunition Data Cards

FSC: 1315

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FEDERAL

FEDERAL SPECIFICATION TT-C-490

Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

TEST METHOD STANDARD No. 151 - Metals; Test Methods

TEST METHOD STANDARD No. 141a

Paint Varnish, Lacquer and Related Materisls; Methods of Inspection, Sampling and Testing

DRAWINGS

ARMY

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
8865054 - Projectile, 81MM, Illuminating, M301A3 Loading Assembly

PUBLICATIONS

ARMY

ET-8865054 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurements functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Code of Federal Regulations, Title 49 - Transportation, Parts 0-190 (The Code of Federal Regulations is available from the Superintendnet of Documents, U.S. Government Printing Office, Washington, D.C., 20402. Orders should specify 49 CFR 0-190 (latest revision) ).

ASTM-A-513 - Specification for Electric-Welded Steel Tubing

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).



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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with all applicable drawings and specifications.

3.2 Heat treatment or stress relief.-Heat treatment or stress relief shall be uniform throughout each lot of body tubes, body tube assemblies, cone tails and tail cone assemblies and shall be so conducted as to produce the physical properties specified on the applicable drawings and in the applicable specifications.

3.2.1 Hot forged heat treat method.-Prior to finish machining, projectile components manufactured from forgings shall be subjected to heat treatment to attain the required heating above the critical temperature, followed by an oil quench. After quenching, the body forging shall be tempered without delay at a temperature suitable to produce required mechancial properties. After removal from the tempering furnace, the body forging shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700°F., after which cooling may be accelerated by water spray.

3.2.2 Cold shaping and hot cup - cold draw methods.-Cold working shall be so accomplished as to produce in the projectile components the mechanical properties specified. The projectile components shall be stress relieved at a temperature of 700°F. minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified on the applicable drawing. The temperature and time duration of a process anneal prior to final forming operation shall be established by the ability to meet the required mechanical properties when formed by an approved sequence of operations.

3.3 Physical properties.-Physical properties required by the applicable drawings or specifications shall be uniform throughout each component within the specified tolerances.

3.3.1 Body Tubes (see dwg. 9232484).-Welded tubes which are to be used for body tubes shall not rupture, or indicate flaws as evidenced by leaks, or undergo permanent deformation when tested as specified in 4.4.7.



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3.4 Metal defects.-All metal components shall be free from cracks, splits, cold shuts, inclusions, porosity or other similar defects.

3.5 Loading assembly.-The projectile loading assembly shall comply with all requirements specified on Drawing (dwg) 8865054 and with all requirements specified in the applicable specifications.

3.6 Moisture

3.6.1 First fire composition.-The moisture content of the first fire composition shall not exceed 0.30 percent, when determined as specified in 4.4.2.1 or 4.4.2.3.

3.6.2 Illuminant composition.-The moisture content of the illuminant composition shall not exceed 0.30 percent, when determined as specified in 4.4.2.2 or 4.4.2.3.

3.7 Burning time.-Candle power of the illuminant assembly (dwg. 8865056).-The illuminant assembly during static testing shall not have a burning time of less than 60 seconds minimum (min). The luminosity of all the illuminant assemblies shall not have an average candle power of less than 500,000 adjusted (min.) and no one particular assembly shall have a luminosity of less than 450,000 adjusted (min.) candle power.

3.8 Proving ground

3.8.1 The projectile loading assembly shall comply with the following requirements when tested as specified in 4.4.8. (The requirements of 3.8.2, 3.8.3, 4.2e, 4.2f, 4.2.3 and 4.2.4 are not applicable when the contractor for the projectile assembly is also the contractor for the assembling and packing of the complete round).

3.8.2 Functioning.-The projectile loading assembly shall function satisfactoryily (see 6.9) and shall provide effective illumination (see 6.10) for 60 seconds, min.

3.8.3 Security of parts.-There shall be no metal parts failure in the bore of the weapon or in the first approximate 200 yards of horizontal range.



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3.9 Workmanship

3.9.1 All parts and assemblies shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.9.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

3.9.3 Surface finish.-The requirements for surface finish shall be as specified on the applicable drawings.

3.9.4 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of Specification MIL-A-2550.

3.9.5 Cloth.-All cloth shall be free of tears, rips and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears or foreign matter.

3.10 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Push test of shear pins (dwg. 8865054).-The tops of the shear pins shall be flush to 0.03 inch below flush after being subjected to the test specified in 4.4.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other



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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Submission of product.*-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review. Additionally, a certificate of conformance denoting static functioning and moisture content of illuminant shall be furnished to the QAR with each lot submitted for acceptance, including statement of findings for candlepower, moisture content and formulation per dwg. 8865059.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 *Submission of parts and assemblies.*-The contractor shall submit a first article sample as designated by the contracting officer for determination of conformance to drawing and specification requirements. The first article sample shall consist of the following:



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a. Fifteen each of every internal part and internal subassembly including loaded illuminant loading assemblies.
b. Ten each of every external metal part and every external metal part subassembly unpainted.
c. Five empty projectile loading assemblies pinned and unpainted (i.e., without internal components).
d. Five empty projectile loading assemblies pinned, painted and marked (i.e., without internal components).
e. Twenty-five painted and pinned projectile loading assemblies assembled with inert loaded illuminant loading assemblies (suggested filler: sand and laminac).
f. Fifteen completely loaded projectile loading assemblies pinned, painted and marked.

The parts and assemblies shall have been produced by the contractor, or furnished by a supplier, using the same production processes, procedures and equipment (except for the inerted candles) as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statement of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government).

4.2.1.1 Salt Spray Inspection (Inorganic finish).-Five samples of each of the following components (dwg. 8865015, 8865017, 8865021, 8865029, 8865030, and 8865034) shall be subjected to a salt spray test for 96 hours (except where otherwise specified on the component drawings) in accordance with the procedure specified in 4.4.5. After exposure, specimens shall be examined and compared with the unexposed specimens for the effect of corrosion. Any evidence of corrosion on the test specimens shall be cause for rejection.



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4.2.1.2 Salt Spray Inspection (Organic Finish).-Five samples of each of the following components and subassemblies (dwg. 9232482 and 9232484 assembled with 9232483) shall be subjected to a salt spray test as specified on drawing 9232481 in accordance with the procedure specified in 4.4.5.1. After exposure, specimens shall be examined and the criteria for acceptance shall be as specified in Federal Specification TT-C-490, paragraph 3.9.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings. In addition, the twenty-five inert loaded projectile loading assemblies and fifteen loaded illuminant assemblies will be subjected to first article ballistic testing as specified in paragraphs 4.2.3.1 and 4.2.3.2 of this specification.

4.2.3 First Article Ballistic Testing

4.2.3.1 Metal parts security.-Twenty-five (25) of the projectile metal parts shall be tested as shown in the phases in Table I. All of the projectiles shall be inert loaded to service weight and will be fired from a weapon for which the projectile is standard. The propellant charge will be adjusted to obtain approximately 8400 pounds per square inch. Dummy or inert fuzes shall be used on all of the metal parts security and recovery test rounds. Pressure data shall be taken for each round fired.

TABLE I

| Phase | Mortar Elevation (deg) | Initial Production Sample |
|---|---|---|
| Security | 45 | 10 |
| Recovery | Optional | 15 |

Recovery.-Rounds fired for recovery which satisfy the requirements of 3.8.3 are for information only to determine quality of ammunition produced. Examination shall be made after recovery for evidence of body or tail cone deformation, dishing or breakup of the projectile other than by impact.



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4.2.3.2 Projectile functioning.-After completion of the first article tests as well as the metal parts security test the projectile functioning tests shall be performed. The fifteen loaded projectiles shall be inspected for conformance to any or all of the requirements of this specification and any or all of the requirements of the applicable drawings at the contractor's plant prior to submission to the proving ground for proving ground testing. Seven of the sample projectiles shall be fired at the Charge 4 zone while the remaining eight will be fired at the Charge 8 zone.

4.2.4 Recovery.-The proving ground shall make the necessary arrangements to recover all of the rounds fired for recovery during the metal parts security test and any or all malfunctioning parachute and illuminant assemblies which may occur during projectile functioning test. Arrangements for the recovery of malfunctions shall be for the first article samples only unless directed by the procuring activity.

4.2.5 Rejection.-If any internal part, inert or loaded, or external metal parts or assemblies fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. If during the metal part security or recovery test any projectile fails to comply with the requirement of 3.8.3 the first article sample shall be rejected. In addition, if during the projectile functioning test any loaded projectile fails to comply with the requirements of 3.8.2 or 3.8.3, the first article projectile functioning sample shall be rejected. The Government reserves the right to terminate its inspection and testing upon any failure in the first article sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a complete first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units



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from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required shall be in accordance with MIL-STD-1168. Each inspection lot of projectile loading assemblies shall contain:

a. Projectile external metal components from one lot interfix number from one manufacturer.

b. Parachute assemblies from one lot interfix number from one manufacturer.

c. Canister assemblies from one manufacturer, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Heat treatment lots.-Heat treatment lots shall contain components from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of tensile samples as described in 4.3.3.7.

4.3.1.2 Stress relieved lots.-Stress relieved lots shall contain cold worked components from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted, shall be the basis for the selection of tensile samples as described in 4.3.3.7.

4.3.1.3 Miscellaneous lots.-Small lots (such as components manufactured from abnormally small heats of steel that have become separated from previously delivered lots, etc.,) may at the discretion of the supplier, with the approval of the Government be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.3.1.1 or 4.3.1.2.



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4.3.1.4 Projectile metal parts lot.-When it has been determined that components of a heat-treated stress relieved or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawing, they may be re-grouped for assembly into projectile lots, to which will be assigned a projectile lot number.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cone, tail (prior to paint), (see dwg. 9232482 covering a detail of dwg. 9232481.

| Categories | Defects | Method of Inspection | Code No. (see 6.3) |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Concentricity of pitch diameter of fin thread with forward outside diameter, including perpendicularity of rear face | Gage | 01001 |
| 102. | Concentricity of bourrelet diameter with forward outside diameter | Gage | 01002 |
| 103. | Perpendicularity of body seat with outside forward diameter | Gage | 01003 |
| 104. | Diameter outside, forward | Gage | 01004 |
| 105. | Diameter inside max, forward | Gage | 01005 |
| 106. | Length of forward diameter, outside | Gage | 01006 |
| 107. | Metal defective | Visual | 01007 |
| 108. | Concentricity of rear outside diameter of cone with forward outside diameter | Gage | 01008 |
| 109. | Concentricity of datum on center of outside taper with forward | | |



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| | | | |
|---|---|---|---|
| Minor: | | AQL 4.0 percent | |
| 201. | Diameter, small outside at rear | Gage | 01010 |
| 202. | Depth of undercut forward of body seat, max | Gage | 01011 |
| 203. | Length from rear face to start of taper | Gage | 01012 |
| 204. | Length total (distance from rear shoulder to front face) | Gage | 01013 |
| 205. | Thickness through base to rear face | Gage | 01014 |
| 206. | Thickness of wall | Gage | 01015 |
| 207. | Start of taper from front shoulder | Gage | 01016 |
| 208. | Width of undercut forward of body seat, max (if applicable) | Gage | 01017 |
| 209. | Surface finish, improper | Visual (see 6.2) | 01018 |
| 210. | Radii or chamfer missing or incorrect | Visual (see 6.2) | 01019 |
| 211. | Evidence of poor workmanship, (see 3.9) | Visual | 01020 |

4.3.2.2 *Cone tail (after paint) (see dwg. 9232482 covering a detail of dwg. 9232481).*

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Pitch diameter of fin thread, min | Gage | 02001 |
| 102. | Major diameter of fin thread, min | Gage | 02002 |
| 103. | Diameter of bourrelet | Gage | 02003 |
| Minor: | AQL 2.5 percent | | |
| 201. | Length of effective fin thread, min | Gage | 02004 |
| 202. | Silicone compound missing, threads | Visual | 02005 |
| 203. | Silicone compound missing on forward outside diameter | Visual | 02006 |
| 204. | Paint on forward outside diameter | Visual | 02007 |
| 205. | Forward outside diameter damaged | Visual | 02008 |
| 206. | Threads damaged or not full | Visual | 02009 |
| 207. | Evidence of poor workmanship, (see 3.9) | Visual | 02010 |



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4.3.2.3 Pin, Groove (see dwg. 9232485 covering a detail of dwg. 9232481).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Diameter expanded | Gage | 03001 |
| Minor: | AQL 1.5 percent | | |
| 201. | Length total | Gage | 03002 |
| 202. | Chamfer missing or incorrect | Visual (see 6.2) | 03003 |
| 203. | Surface finish, improper | Visual (see 6.2) | 03004 |
| 204. | Evidence of poor workmanship, (see 3.9) | Visual | 03005 |

4.3.2.4 Adapter, (see dwg. 9232483 covering a detail of dwg. 9232481).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Concentricity of pitch diameter threads with diameter at rear shoulder including perpendicularity with front face | Gage | 04001 |
| 102. | Parallelism of shoulder with front face | Gage | 04002 |
| 103. | Diameter at rear shoulder | Gage | 04003 |
| 104. | Diameter of shoulder | Gage | 04004 |
| 105. | Metal defective | Visual | 04005 |
| Minor: | AQL 2.5 percent | | |
| 201. | Diameter at front | Gage | 04006 |
| 202. | Length from front face to shoulder | Gage | 04007 |
| 203. | Length total | Gage | 04008 |
| 204. | Staking notches missing or incorrect (see 6.2) | Visual | 04009 |
| 205. | Chamfer missing or incorrect (see 6.2) | Visual | 04010 |
| 206. | Surface finish improper (see 6.2) | Visual | 04011 |
| 207. | Evidence of poor workmanship, (see 3.9) | Visual | 04012 |



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4.3.2.5 Tube Body (see dwg. 9232484 covering a detail of dwg. 9232481).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.5 percent | | |
| 101. | Front face not parallel with rear face | Gage | 05001 |
| 102. | Diameter of adapter cavity | Gage | 05002 |
| 103. | Diameter of bourrelet, min | Gage | 05003 |
| 104. | Diameter of charge cavity | Gage | 05004 |
| 105. | Diameter of cone cavity | Gage | 05005 |
| 106. | Diameter of body outside | Gage | 05006 |
| 107. | Thickness of wall, min | Gage | 05007 |
| 108. | Metal defective | Visual | 05008 |
| 109. | Concentricity of outside diameter with cone cavity | Gage | 05009 |
| 110. | Concentricity of bourrelet with cone cavity including perpendicularity with back face | Gage | 05010 |
| 111. | Concentricity of adapter cavity with cone cavity including perpendicularity with back face | Gage | 05011 |
| Minor: | AQL 2.5 percent | | |
| 201. | Depth of adapter cavity | Gage | 05012 |
| 202. | Depth of cone cavity | Gage | 05013 |
| 203. | Length of bourrelet | Gage | 05014 |
| 204. | Length, total | Gage | 05015 |
| 205. | Surface finish, improper | Visual (see 6.2) | 05016 |
| 206. | Radii or chamfers missing or incorrect | Visual (see 6.2) | 05017 |
| 207. | Evidence of poor workmanship, (see 3.9) | Visual | 05018 |



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4.3.2.6 Metal parts assembly (prior to painting) (see dwg. 9232481).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Concentricity of outside diameter with adapter thread including perpendicularity of front face | Gage | 06001 |
| 102. | Length total, max | Gage | 06002 |
| 103. | Pin(s) missing | Visual | 06003 |
| 104. | Pin(s) not flush or below flush | Visual | 06004 |
| 105. | Distance front face adapter to shear pin holes | Gage | 06005 |
| Minor: | AQL 2.5 percent | | |
| 201. | Diameter pin hole (prior to pinning) | Gage | 06006 |
| 202. | Depth of pin hole (prior to pinning) | Gage | 06007 |
| 203. | Location of pins circumferential, (see 6.2) | Visual | 06008 |
| 204. | Radii or chamfer missing or incorrect (see 6.2) | Visual | 06009 |
| 205. | Marking missing, incorrect or illegible | Visual | 06010 |
| 206. | Evidence of poor workmanship (see 3.9) | Visual | 06011 |

4.3.2.7 Metal Parts Assembly (after painting) (see dwg. 9232481).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Pitch diameter of thread, max | Gage | 07001 |
| 102. | Minor diameter of thread, max | Gage | 07002 |
| 103. | Diameter of bourrelet, max | Gage | 07003 |
| Minor: | AQL 2.5 percent | | |
| 201. | Pitch diameter of thread, min | Gage | 07004 |
| 202. | Interior paint, inadequate | Visual | 07005 |
| 203. | Exterior paint, inadequate or damaged | Visual | 07006 |
| 204. | Paint on rear inside diameter | Visual | 07007 |
| 205. | Silicone compound missing, rear inside diameter | Visual | 07008 |
| 206. | Silicone compound missing, threads | Visual | 07009 |



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4.3.2.8 Spacer, parachute (see dwg. 8865034 covering a detail of dwg. 8865054).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: None defined. | | | |
| Minor: | AQL 1.5 percent | | |
| 201. | Total length, minimum (min)..... | Gage | 08001 |
| 202. | Thickness..... | Gage | 08002 |
| 203. | Evidence of poor workmanship (see 3.9)..... | Visual | 08003 |

4.3.2.9 Cup, parachute (see dwg. 8865010 covering a detail of dwg. 8865054).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: None defined. | | | |
| Minor: | AQL 1.00 percent | | |
| 201. | Height..... | Gage | 09001 |
| 202. | Outside diameter, min..... | Gage | 09002 |

4.3.2.10 Disc, coupling (see dwg. 8865021 covering a detail of dwg. 8865054).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Outside diameter, min..... | Gage | 10001 |
| Minor: | AQL 1.50 percent | | |
| 201. | Thickness..... | Gage | 10002 |
| 202. | Center hole diameter..... | Gage | 10003 |
| 203. | Any operation missing..... | Visual | 10004 |



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4.3.2.11 Parachute holder assembly (see dwg. 8865032 covering a detail of dwg. 8865046).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Tab missing | Visual | 11001 |
| 103. | Eyelets missing or improperly crimped | Visual | 11002 |
| 103. | Pull cord improperly attached | Visual | 11003 |
| Minor: | AQL 1.0 percent | | |
| 201. | Stitching broken | Visual | 11004 |
| 202. | Evidence of poor workmanship, (see 3.9) | Visual | 11005 |

4.3.2.12 Parachute assembly, prior to inserting in parachute holder (see dwg. 8865046 and 8865057).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Holes or tears in canopy | Visual | 12001 |
| 102. | Clamp or tying cord missing or improperly secured to shroud lines | Visual | 12002 |
| 103. | One or more shroud lines or center cord missing | Visual | 12003 |
| 104. | Unpatched holes of more than six broken picks | Visual | 12004 |
| Minor: | AQL 2.5 percent | | |
| 201. | Developed diameter of canopy | Gage | 12005 |
| 202. | Length of shroud lines from canopy hem to tying cord or clamp | Gage | 12006 |
| 203. | Stitching missing or broken | Visual | 12007 |
| 204. | Crown missing | Visual | 12008 |
| 205. | Marking misleading or unidentifiable | Visual | 12009 |
| 206. | Length of center cord to tying cord or clamp | Gage | 12010 |
| 207. | Evidence of poor workmanship, (see 3.9) | Visual | 12011 |



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4.3.2.13 Parachute holder assembly, after insertion of parachute assembly (see dwg. 8865046 covering a detail of dwg. 8865054) *

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Improper number of folds first operation | Visual | 13001 |
| 102. | Improper number of accordian folds fifth operation | Visual | 13002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship, (see 3.9) | Visual | 13003 |

*NOTE: These inspections are to be performed on completed lots of parachute assemblies. After inspection the parachutes may be refolded, inserted in the parachute holder assembly and returned to the lot.

4.3.2.14 Illuminant Assembly (see dwg. 8865056 covering a detail of dwg. 8865054).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Any component missing or improperly assembled | Visual | 14001 |
| 102. | First fire missing or disc improperly assembled | Visual | 14002 |
| 103. | Quick match assembly improper | Visual | 14003 |
| Minor: | AQL 1.5 percent | | |
| 201. | Rivets above flush of outer contour | Visual | 14004 |
| 202. | Loop improper | Visual | 14005 |
| 203. | Marking incomplete, misleading or unidentifiable | Visual | 14006 |
| 204. | Evidence of poor workmanship, (see 3.9) | Visual | 14007 |



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4.3.2.15 Loading assembly prior to assembly of tail cone and protective plugs (see dwg. 8865054).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Shroud line not attached or improperly assembled to suspension loop | Visual | 15001 |
| 102. | Pull out cord improperly attached to coupling disc | Visual | 15002 |
| 103. | Any component missing or damaged | Visual | 15003 |
| Minor: | AQL 1.5 percent | | |
| 201. | Illuminating assembly improperly seated | Visual | 15004 |
| 202. | Disc missing or improperly installed | Visual | 15005 |
| 203. | Disc parachute missing or improperly installed | Visual | 15006 |
| 204. | Evidence of poor workmanship, (see 3.9) | Visual | 15007 |

4.3.2.16 Loading, Assembly, after assembly of shear pins (see dwg. 8865054).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Body or tail cone split | Visual | 16001 |
| 2. | One or more shear pins missing | Visual | 16002 |
| Major: | AQL 1.0 percent | | |
| 101. | Tail cone improperly assembled to body | Visual | 16003 |
| 102. | Any shear pin above flush | Visual | 16004 |
| 103. | Depth of shear pin (max) | Gage | 16005 |
| 104. | Location of shear pins | Gage | 16006 |
| 105. | Excessive gap between tail cone and body tube | Gage | 16007 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 1.5 percent | | |
| 201. | Protective dust plug missing or loose..... | Visual/ Manual | 16009 |
| 202. | Marking misleading or unidentifiable..... | Visual | 16010 |
| 203. | Evidence of poor workmanship, (see 3.9)..... | Visual | 16011 |

4.3.2.17 Parachute holder assembly (see dwg. 9242685 covering a detail of dwg. 8865046). (When Applicable)

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Tab missing..... | Visual | 17001 |
| 102. | Eyelits missing or improperly crimped..... | Visual | 17002 |
| 103. | Pull cord improperly attached..... | Visual | 17003 |
| 104. | Inside diameter of bag..... | Gage | 17004 |
| Minor: | AQL 1.0 percent | | |
| 201. | Stitching broken..... | Visual | 17005 |
| 202. | Evidence of poor workmanship, (see 3.9)..... | Visual | 17006 |

4.3.2.18 Parachute assembly prior to inserting in parachute holder (see dwg. 9242684 covering a detail of dwg. 8865046).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Clamp or tying cord missing or improperly secured to shroud lines.. | Visual | 18001 |
| 102. | One or more shroud lines or center cord missing..... | Visual | 18002 |
| 103. | Unpatched holes of more than six broken picks..... | Visual | 18003 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 2.5 percent | | |
| 201. | Developed diameter of canopy | Gage | 18004 |
| 202. | Length of shroud lines from canopy hem to tying cord or clamp | Gage | 18005 |
| 203. | Stitching missing or broken | Visual | 18006 |
| 204. | Marking misleading or unidentifiable | Visual | 18007 |
| 205. | Length of center cord to tying cord or clamp | Gage | 18008 |
| 206. | Evidence of poor workmanship, (see 3.9) | Visual | 18009 |

4.3.3 Testing

4.3.3.1 Static functioning (see 3.7) - Major defect - Code No. 14008.-The test shall be performed as specified in 4.4.1. A sample of three illuminant canister assemblies shall be selected out of each batch of candles (see 6.4). Two canister assemblies shall be tested for static burning time and if any burning time is less than 60 seconds or any candlepower is less than 450,000 adjusted the batch shall be rejected. If the average candle power is less than 500,000 adjusted, the third sample shall be tested for static burning time, and if the burning time is less than 60 seconds for the third sample or the average candlepower of the three samples is less than 500,000 adjusted the batch represented shall be rejected.

4.3.3.2 Moisture content

4.3.3.2.1 First fire composition - Major defect - Code No. 14009.-The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce first fire sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirements, the eight hours of production represented by the sample shall be rejected.

4.3.3.2.2 Illuminant composition - Major defect - Code No. 14010.-The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce illuminant sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the application requirements, the eight hours of production represented by the sample shall be rejected.



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4.3.3.3 Pull test of illuminant assembly (see dwg. 8865056) - Major defect - Code No. 14011.-Fifty samples shall be selected from each lot for this test. If any assembly visibly yields under the load the lot shall be rejected.

4.3.3.4 Non-Destruct Electrical test of body tubes (see dwg. 8232484) - Critical defect - Code No. 05019.-All welded tubes that are to be used for body tubes shall be subjected 100 percent to the non-destructive electrical test cited in ASTM-A-513 paragraph S7 (see 6.6).

4.3.3.5 Hydrostatic Test of Body Tubes (see dwg. 9232484) - Critical defect - Code No. 05020.-All welded tubes that are to be used for body tubes shall be subjected to the hydrostatic test specified in 4.4.7 (see 6.6).

4.3.3.6 Profile and alignment (see dwg. 8865054) - Major defect - Code No. 16013.-The loading assembly shall be gaged 100 percent for profile and alignment. Any loading assembly which fails to comply with the requirement specified shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.3.4. This requirement is not applicable when the contractor for the loading assembly is also the contractor for the complete round.

4.3.3.7 Mechanical properties

4.3.3.7.1 Tension test

4.3.3.7.1.1 Body Tube (see dwg. 9232484)-Major defect-Code No. 05021.-After final heat treatment or stress relief, two body tubes shall be selected from each heat treated or stress relieved lot of body tubes and subjected to the tension test to determine compliance with the yield strength and elongation requirements of the component drawing. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.7.1.2 Cone, tail (see dwg. 9232482)-Major defect-Code No. 01021.-After final heat treatment or stress relief, two tail cones shall be selected from each heat treated or stress relieved lot of tail cones and subjected to the tension test to determine compliance with the yield strength and elongation requirements of the component drawing. Failure of any specimen shall be cause for rejection of the lot.



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4.3.3.7.1.3 Adapter (see dwg. 9232483) - Major defect - Code No. 04013.-The sample size and acceptance criteria shall be as specified in the applicable material specification, except that the mechanical properties specified on the applicable component drawing shall apply to the finished components. Failure to comply with the requirements of the applicable drawing and specifications shall be cause for rejection of the lot.

4.3.3.7.1.4 Miscellaneous lots.-Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.3.3.7.1.1 or 4.3.3.7.1.2 and 4.3.3.7.1.3, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Double the number of samples for regular lots shall be selected for miscellaneous lots.

4.3.3.8 Proving ground (see 3.8).-Defects shall be classified in accordance with the following table: The test shall be performed as specified in 4.4.8.

| Categories | Defects | Code No. |
|---|---|---|
| Critical: | Premature burst or metal parts separation in first approximate 200 yards horizontal range (see 3.8.3) | 16014 |
| Major: | Tail cone and body tube assembly fail to separate (see 3.8.2) | 16015 |
| Major: | Illuminant fails to ignite, or provides effective illumination for 40.0 seconds or less (see 3.8.2) | 16016 |
| Minor: | Effective illumination for more than 40.0 seconds but less than 60.0 seconds (see 3.8.2) | 16017 |



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4.3.3.8.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified herein, a sample of 80 loaded assemblies shall be randomly selected from each lot for this test. If any critical defect occurs; if a total of 6 or more Major defects occur, or if 8 or more Minor defects occur, the lot shall be rejected.

4.3.3.8.2 After three consecutive lots have met the criteria of 4.3.3.8.1, a sample of 32 loaded assemblies shall be randomly selected from each lot for this test. If any critical defect occurs, if a total of 3 or more major defects occur, or if 4 or more minor defects occur, the lot shall be rejected. In the event that three successive lots fail to meet the requirements of 4.3.3.8.2, the sampling provisions of 4.3.3.8.1 shall again apply. When three consecutive lots have again successfully passed the requirements contained therein the provisions of 4.3.3.8.2 shall be reinstated.

4.3.3.9 Push test of shear pins (see 3.11) - Critical defect - Code No. 16018.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter M (Single Sampling), utilizing an AQL of 0.04 percent. If any loading assembly fails to comply with push test requirements specified in 3.11, the lot of loading assemblies shall be rejected.

4.3.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-8865054 identifies the inspection equipment required to perform the examinations and tests prescribed in this specification. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.5.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provide and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.



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4.3.4.2 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Static functioning test.-In calculating the candle power of any one illuminant assembly the best continuous 60 seconds are to be used if the total burning time is greater than 60 seconds. The static test shall be conducted in accordance with the procedures and equipment specified in paragraph 4.3.4. The distance from the photometric cell to the flare shall be 60.00 plus or minus .25 ft. The axial center of the photometric cell and the axial center of the fireball shall be located 54.0 plus or minus .5 inch from the floor. The temporary backwall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is 15 ft. The value recorded for candlepower for each candle tested shall be adjusted by adding 225,000 candles. The adjusted values shall be used to determine compliance with the requirements specified. The center line of baffle opening shall be in line with axial center of fire ball and axial center of photometric cell.

4.4.2 Moisture

4.4.2.1 First fire composition.-The moisture content of the first fire composition shall be determined in accordance with Standard MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of dimethylformamidemethyl alcohol and anhydrous methanol.

4.4.2.2 Illuminant composition.-The moisture content of the illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in Standard MIL-STD-1234, Method 101.2 up to paragraph 5.3. A 50 ml. sample of 1:1 dimethylformamidemethyl alcohol solution shall be added into a 250 ml. Erlenmeyer flask (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered



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and shaken vigorously to thoroughly mix solution and pyrotechnic sample. Allow the solution to set 5 to 15 minutes. A blank sample of the 1:1 dimethylformamide methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into the dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.5 - 0.25 minutes. 25 ml. of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernantant liquid into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide methanol blank solution shall be conducted as spedified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation:

$$\frac{AB}{5\ (10)} = \text{Percent } H_2O$$

Less STd. Blank

Leaves Percent Moisture in Sample

Where:

A = Mls of Karl Fischer to titrate sample
B = Standardization Factor (mg./ml.)
10 = Constant Factor
5 = Standard Sample Weight

4.4.2.3 Alternate method for first fire composition moisture and illuminant composition moisture.-The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4, Standard MIL-STD-286. The temperature setting shall be 100 degrees Centigrade and the time setting 30 minutes. The sample size shall be between 0.40 and 0.60 grams.



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4.4.3 Pull test of illuminant assembly.-The illuminant assembly shall be tested by supporting each assembly in a suitable fixture and applying a steady pull to the suspension wire assembly by inserting a hook or pin through the loop. The oval sleeve, suspension wire assembly and suspension cup shall withstand the load for a minimum of three seconds.

4.4.4 Parachute Inspections.-Parachute assembly (see dwg. 8865057).-Shroud lines and center cord shall be inspected under a 1/2 plus or minus .01 pound tensile load to determine conformance to the drawing requirement. The distance across the points of the gores shall be inspected by laying out the completed parachute on a plane surface and measuring across all the gore points while all shroud lines are under a 1/2 plus or minus .01 pound tensile load to determine conformance to drawing requirements.

4.4.5 Salt spray test.-Inert component parts or sub-assemblies selected as in 4.2.1.1 shall be subjected to a 5 percent salt spray test (except where otherwise specified on the drawing) in accordance with FED-STD-151, Method 811.1.

4.4.5.1 Salt Spray Test.-Samples selected in accordance with 4.2.1.2 shall be tested in accordance with FED-STD-141a, method 6061. Prior to performing the test, the samples shall be scored in accordance with the procedure specified in Federal Specification TT-C-490, paragraph 4.2.8.

4.4.6 Mechanical properties

4.4.6.1 Body tube.-From each sample selected in accordance with 4.3.3.7.1.1 or 4.3.3.7.1.4, two tension test specimens conforming to the Type T-1 prescribed by Federal Test Methods Standard No. 151 shall be taken longitudinally 180 degrees apart. Yield strength shall be determined by either the offset method or the extension under load method as prescribed by Federal Test Method Standard No. 151. The limiting set shall be 0.1 percent (0.001 inches per inch of gage length).

4.4.6.2 Cone, tail.-From each sample selected in accordance with 4.3.3.7.1.2 or 4.3.3.7.1.4, two tension test specimens of the largest obtainable R type specimen conforming to the prescribed Federal Test Method Standard No. 151 shall be taken longitudinally from the straight portion of the component, prior to machining, 180 degrees apart. Yield stregnth shall be determined by either the offset method or the extension under load method as prescribed by Federal Test Method Standard No. 151. The limiting set shall be 0.1 percent (0.001 inches per inch of gage length).



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4.4.7 Hydrostatic test of body tubes.-All welded tubes that are to be used for body tubes shall withstand the internal hydrostatic pressure as calculated by the following formula. However, the pressure applied to the tube wall shall be 5,000 pounds per square inch if the formula parameters dictate the use of pressure greater than this value:

$$P = \frac{1.6\ St}{D}$$

Where:
P = Test pressure in pounds per square inch
S - Minimum specified yield strength, psi
t = specified minimum wall thickness, in inches
D = Specified maximum outside diameter of tube in inches

Any tube which does not sustain the hydrostatic test pressure for a minimum of 10 seconds or evidences a permanent deformation after hydrostatic test shall be rejected. Equipment used shall be in accordance with 4.3.4.

4.4.8 Proving ground.-The test shall be performed at a Government Proving ground in accordance with the applicable Acceptance Test Procedures.

4.4.8.1 Functioning.-The projectile assembly shall be assembled with accepted fuzes, fin assemblies, adapters, ignition cartridges, primers and propellant increments and fired from a 81MM Mortar for which the round is standard. One half of the sample cartridges shall be fired with an ignition cartridge and four propellant increments (charge zone four) and one half of the sample cartridges shall be fired with an ignition cartridge and eight propellant increments (charge zone eight) and in accordance with elevation specified in Table II. However, if meterological conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to increase the height of burst and insure effective illumination. Observation shall be made for compliance with the requirements specified for security of parts and "effective illumination" (see 6.10). In addition, muzzle velocity height of burst and time of fuze functioning shall be recorded for information only for each round tested. Any malfunction occurring during the test shall be reported in the official firing record. Meterological data shall also be reported in the official firing record.



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TABLE II

| Charge Zone | Fuze Setting | Mortar Elevation (degrees) |
|---|---|---|
| 4 | 20 sec | 60 |
| 8 | 30 sec | 60 |

NOTE: If a projectile functions properly in all respects, with the exception that the "effective illumination" time is less than sixty (60) seconds because of illuminant and parachute assembly hit the ground while still burning and if the burst height, that is, the height above ground level at which the tail cone separated from the tube body was less than 1,500 feet the performance of the projectile will not be considered in evaluating the acceptability of the lot.

4.4.8.2 Recovery of malfunctioning first article samples.-The proving ground shall make the necessary arrangements to recover any or all malfunctioning parachute and illuminant assemblies which may occur during the first article ballistic test. Recovery of malfunctions shall be for the first article samples only unless otherwise directed.

4.4.9 Push test of shear pins.-An axial load of 100 plus/minus 5 pounds shall be applied to the top of each shear pin using test equipment in accordance with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-Metal parts assemblies shall be packed in accordance with MIL-STD-1169.

5.1.2 Level C.-The projectile loading assembly shall be packed in accordance with Code of Federal Regulations 49 CFR 0-190 in a manner that will afford adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity.

5.2 Labeling and marking

5.2.1 Labeling and marking of packed metal parts assemblies shall be in accordance with MIL-STD-1169.



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5.2.2 Marking.-Each shipping container shall be marked in accordance with dwg. 8796522 for interplant shipment. The FSC number and DOD identification codes are not applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Requirements for the submission of first article samples by the contractor.

6.2 Visual Examination Qualification.-When compliance with applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor.

6.4 Batch.-A batch is defined as the quantity of material which has been manufactured by some unit chemical process and subjected to some physical operation intended to make the final product substantially uniform. However, for the purposes of this specification a batch of candles shall be defined as that quantity of candles which has been loaded with one batch of illuminant using the one batch of illuminant for all increments.

6.5 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.5.1 through 6.5.5.

6.5.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawing or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.5.2. Design action required of the contractor is described in paragraphs 6.5.3 and 6.5.4. The contractor will be required to prepare detailed drawings in accordance with 6.5.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.5.5.



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6.5.2 Inspection equipment list codes.-The inspection equipment as defined in 6.5.3 and 6.5.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army design, mandatory for use

6.5.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.5.5.

6.5.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.5.5.



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6.5.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny ARsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.6 If the hydrostatic and non-destructive electrical tests have been performed by either the mill supplier or metal parts vendor, and the tubing has been found to conform to specification requirements, a second hydrostatic or non-destructive electrical test need not be performed.

6.7 If the mill lot of body tubing does not undergo any stress-relieving or heat-treating during fabricating into final body tubes (see dwg. 9232484), the specimens for the tensile, yield and elongation tests may be taken from mill tubes from the lot of tubing rather than from the component tube body.

6.8 Static burning time.-Minimum static burning time as referenced herein (see 3.7) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee final ballistic minimum burning time requirement. Structual integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirement and is attained by proper formulation, blending, consolidation and cure.

6.9 Satisfactory functioning.-Satisfactory functioning shall be defined as ejection of the parachute and illuminant assembly and proper suspension and ignition of the illuminant such as to assure "effective illumination".



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6.10 Effective illumination.-Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

6.11 Statements of findings shall show the specific analytical results of each test or inspection performed as required by the applicable drawing or specification together with the specified acceptable limits.

6.12 Submission of test data.-In addition to the normal distribution of records, when the projectile is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2 and DA-4, Dover, New Jersey, 07801.

Custodian:
ARMY-MU

Preparing Activity:
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Project Number : 1315-A-568



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